Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 1 / 方法 1 [NT06-M01]

Expand the numeral before manipulating it, and distinguish digit values from their place weights. Digit bounds, leading zeros and carry behavior belong in the model from the first line.

运算前先展开数码，区分数字值与位权。数位范围、首位零与进位行为从建模第一行就应明确。

Expand place values and enforce the common digit range

展开位值并施加共同数位范围

Procedure

1. Write a base-b numeral as $\sum d_i b^i$, with fractional positions using negative powers.

2. A digit shared between bases must be valid in every base in which it occurs; each leading digit is nonzero.

3. Eliminate variables, use a small congruence or a power bound, then test the remaining digits exactly.

4. When all digits are nonnegative, compare the value monotonically as the base changes.

通用步骤

1. 把 b 进制数写成 $\sum d_i b^i$，小数位使用负指数。

2. 跨进位制共用数字须在每个出现的进位制中有效，且每个首位非零。

3. 消元后使用小模数或幂的范围界，再精确检查剩余数字。

4. 所有数位非负时，可按底数变化单调比较数值。

> Example. The numeral $\overline{1d2}_b$ has value $2b^2$ for an integer base $b > 2$. Find all valid (b, d).

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 1 / 方法 1 [NT06-M01] (continued / 续)

> 例题：整数底数 $b > 2$ 下，数码 $\overline{1d2}_b$ 的值为 $2b^2$。求全部有效 (b, d)。

Worked solution

1. Place value gives $b^2 + db + 2 = 2b^2$, or $b(b - d) = 2$.

2. Because $0 \le d < b$, the integer $b - d$ is positive, so b must divide 2.

3. This contradicts $b > 2$. There are no valid digit/base pairs.

例题解析

1. 位值给出 $b^2 + db + 2 = 2b^2$，即 $b(b - d) = 2$。

2. 因 $0 \le d < b$，整数 $b - d$ 为正，故 b 须整除 2。

3. 这与 $b > 2$ 矛盾，故无有效数位底数对。

A decimal-looking symbol is a digit string until its base has been specified.

外观像十进制数的符号，在底数确定前只是数位串。

Practice / 配套练习: AIME-2001-I-Q08 (WB p. 585), AIME-2001-I-Q01 (WB p. 558), AIME-2004-I-Q01 (WB p. 563), AIME-2009-I-Q01 (WB p. 566), AIME-2017-I-Q05 (WB p. 572), AIME-2018-I-Q02 (WB p. 574), AIME-2018-II-Q03 (WB p. 575), AIME-2020-I-Q03 (WB p. 577), AIME-2022-I-Q02 (WB p. 579), AIME-2025-I-Q01 (WB p. 582), AIME-2026-II-Q04 (WB p. 584), AIME-2024-II-Q14 (WB p. 593)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 2 / 方法 2 [NT06-M02]

Isolate a prefix or suffix and reduce digit changes modulo a divisor

隔离前缀或后缀，并对数位变化取模

Procedure

1. For a suffix of length k, write $N = b^k q + r$ with $0 \le r < b^k$.

2. Deletion or replacement changes a known place-value term; reduce the resulting equation or divisibility before enumerating.

3. For a square suffix, solve the congruence for its root modulo prime powers and order the roots, not the suffixes.

4. Check leading zeros, positivity of the remaining prefix and any boundary block separately.

通用步骤

1. 对长度为 k 的后缀，写 $N = b^k q + r$，其中 $0 \le r < b^k$。

2. 删位或替换数位会改变已知位值项；枚举之前先约化所得方程或整除式。

3. 平方数后缀题应按质数幂求根的同余，并给根排序，而非仅给后缀排序。

4. 单独检查首位零、剩余前缀正性及边界区块。

> Example. If $100q + 37$ is divisible by 7, find $q \pmod 7$. Can $100q + 12$ also be divisible by 7?

> 例题：若 $100q + 37$ 可被 7 整除，求 q 模 7 的剩余。$100q + 12$ 也能被 7 整除吗?

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 2 / 方法 2 [NT06-M02] (continued / 续)

Worked solution

1. Modulo seven, $100q + 37 \equiv 2q + 2$, so $q \equiv -1 \equiv 6$.

2. The two integers differ by 25, which is not divisible by seven.

3. Thus they cannot both be divisible by seven. In fact $100q + 12 \equiv 2 \cdot 6 + 5 \equiv 3 \pmod 7$.

例题解析

1. 模七有 $100q + 37 \equiv 2q + 2$，所以 $q \equiv -1 \equiv 6$。

2. 两整数之差为 25，不能被七整除。

3. 因此它们不能同时被七整除；实际 $100q + 12 \equiv 2 \cdot 6 + 5 \equiv 3 \pmod 7$。

A suffix may contain initial zeroes even though the whole numeral may not.

后缀允许开头为零，整个整数的首位则不允许。

Practice / 配套练习: AIME-2000-I-Q01 (WB p. 463), AIME-2006-I-Q03 (WB p. 564), AIME-2012-I-Q10 (WB p. 499), AIME-2014-I-Q08 (WB p. 500), AIME-2020-I-Q04 (WB p. 488), AIME-2024-II-Q07 (WB p. 506)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 3 / 方法 3 [NT06-M03]

Use reversal and palindrome symmetry before counting digits

计数前先使用翻转与回文对称

Procedure

1. Expand a numeral and its reversal; paired places often cancel or acquire equal coefficients.

2. A palindrome is determined by its left half and, for odd length, its center digit.

3. Apply nonzero-leading and digit-sum parity restrictions, then count the independent positions.

4. For averages, average each free digit separately; for two bases, first bound the possible lengths.

通用步骤

1. 展开原数及其翻转，成对位值常会消去或得到相同系数。

2. 回文数由左半部分及奇数长度时的中央数字决定。

3. 施加首位非零与数位和奇偶限制，再统计独立位置。

4. 求平均时分别平均每个自由数字；跨进位制时先限制可能长度。

> Example. How many three-digit decimal palindromes are divisible by 3, and what is their mean?

> 例题：有多少个三位十进制回文数可被 3 整除？它们的平均数是多少？

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 3 / 方法 3 [NT06-M03] (continued / 续)

Worked solution

1. Write $N = 101a + 10b$, with $1 \le a \le 9, 0 \le b \le 9$. Divisibility requires $2a + b \equiv 0 \pmod 3$.

2. For $a \equiv 0$ there are four possible b; for the other two classes there are three. Each a class has three members, giving $3 \cdot 4 + 6 \cdot 3 = 30$.

3. Over these thirty choices, $\sum a = 3(1 + \cdots + 9) + (3 + 6 + 9) = 153$, while the three residue classes give $\sum b = 3(18 + 12 + 15) = 135$. The mean is $(101 \cdot 153 + 10 \cdot 135)/30 = 5601/10$.

例题解析

1. 写 $N = 101a + 10b$，其中 $1 \le a \le 9, 0 \le b \le 9$。整除要求 $2a + b \equiv 0 \pmod 3$。

2. $a \equiv 0$ 时 b 有四种，另两类各三种；每类 a 有三个值，故共 $3 \cdot 4 + 6 \cdot 3 = 30$ 个。

3. 这三十组选取中，$\sum a = 3(1 + \cdots + 9) + (3 + 6 + 9) = 153$，三类剩余给出 $\sum b = 3(18 + 12 + 15) = 135$。平均数为 $(101 \cdot 153 + 10 \cdot 135)/30 = 5601/10$。

> Do not impose distinct digits unless the question says so.

> 题目未要求时，不应额外施加数字互异。

Practice / 配套练习: AIME-2002-I-Q03 (WB p. 560), AIME-2002-II-Q01 (WB p. 561), AIME-2012-I-Q01 (WB p. 568), AIME-2021-II-Q01 (WB p. 794), AIME-2023-II-Q02 (WB p. 580), AIME-2026-I-Q02 (WB p. 583)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 4 / 方法 4 [NT06-M04]

Make carries explicit in column sums and digit-sum changes

明确逐列进位与数位和变化

Procedure

1. Write one equation per column, introducing the carry entering and leaving it.

2. In base b, every carry reduces the naive digit sum by $b-1$; include carries created by incoming carries.

3. Translate fixed totals into constraints on column digit sums, then optimize or count their allowed digit assignments.

4. Verify the complete numeral at the end, including any new leading carry.

通用步骤

1. 每列写一式，设置进入与离开的进位。

2. b 进制中每次进位使原始数位和减少 $b-1$，并须计入由前列进位诱发的进位。

3. 把固定总数转为各列数位和限制，再优化或统计允许数位分配。

4. 末尾检验整个数值，包括新产生的最高位进位。

> Example. For binary integers of a fixed length of at least two bits, characterize when adding one leaves the number of ones unchanged.

> 例题：对位数固定且至少有两位的二进制整数，刻画何时加一后数字一的数目保持不变。

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 4 / 方法 4 [NT06-M04] (continued / 续)

Worked solution

1. If the number ends in t ones, adding one changes those t ones to zeroes and the preceding zero to one.

2. The net change in weight is $1-t$. Equality of weights requires exactly one trailing one.

3. Thus the numeral must end in 01 and become one ending in 10. An all-one string of length at least two has change $1-t<0$, so its overflow cannot qualify.

例题解析

1. 若末尾连续有 t 个一，加一会把它们变成零，并把此前的零变成一。

2. 一的数目净变化为 $1-t$，保持不变要求恰有一个末尾一。

3. 故原数须以 01 结尾，更新后以 10 结尾。至少两位的全一串满足 $1-t<0$，所以溢出不能合格。

Count carry events, not the number of columns whose original digit sum exceeds the base.

应统计实际进位事件，不能只数原数位和超过底数的列。

For $a+b=10^k$, first separate the case of zero units digits. Otherwise the units sum to ten and every higher pair of padded digits sums to nine. A zero in one higher position therefore forces nine in the complementary position, which helps make forbidden-zero cases disjoint. Leading padding zeroes are not digits of the original positive numeral.

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 4 / 方法 4 [NT06-M04] (continued / 续)

For coefficients $0 \le a_i < b^2$, split $a_i = bb_i + c_i$ into legal base-b digits. Then $\sum a_i b^i = bB + C$, with $B = \sum b_i b^i$ and $C = \sum c_i b^i$. Bound both B, C in their k-digit ranges before counting; conversely their padded digit expansions uniquely recover every coefficient. This supplies a reversible digit-stream encoding rather than only a carry count.

对 $a + b = 10^k$，先分出个位均为零的情形；否则个位之和为十，此后每对补齐位数的高位数字之和为九。某个高位为零便迫使互补位置为九，可据此证明某些禁零情形互斥。首部补齐的零不属于原正整数的数位。

若系数满足 $0 \le a_i < b^2$，将其拆为合法 b 进制数字 $a_i = bb_i + c_i$，则 $\sum a_i b^i = bB + C$，其中 $B = \sum b_i b^i, C = \sum c_i b^i$。计数前先给 B, C 都施加 k 位范围界；反过来，二者补齐后的数位展开能唯一恢复每个系数。这提供可逆的数位流编码，而非仅仅统计进位。

Practice / 配套练习: AIME-2006-II-Q07 (WB p. 586), AIME-2012-I-Q05 (WB p. 569), AIME-2015-I-Q08 (WB p. 588), AIME-2016-I-Q08 (WB p. 589), AIME-2019-I-Q01 (WB p. 576), AIME-2024-II-Q03 (WB p. 581), AIME-2010-I-Q10 (WB p. 712)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 5 / 方法 5 [NT06-M05]

Count and unrank digit strings under an upper bound

统计并反向定位上界内的数位串

Procedure

1. Count shorter lengths first, fixing a nonzero leading digit unless zero padding is explicitly allowed.

2. For equal lengths, scan from the leading position and count all allowed smaller-prefix blocks before following the bound.

3. For a specified rank, subtract entire block counts until the desired rank lies in one block, then continue recursively.

4. Use binomial coefficients for a fixed digit weight and handle partial final blocks exactly.

通用步骤

1. 先统计较短长度；除非明确允许补零，首位须非零。

2. 等长时从高位扫描，先统计所有允许的较小前缀块，再沿上界前缀继续。

3. 给定排名时，逐块减去完整块数，直到目标位于某块，再递归处理。

4. 固定数位重数时用二项式系数，并精确处理最后不完整块。

> Example. List the fifth nonnegative binary string of length six with exactly two ones, when strings are ordered by value.

> 例题：六位二进制串（允许前补零）恰有两个一，按数值排序的第五个是什么？

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 5 / 方法 5 [NT06-M05] (continued / 续)

Worked solution

1. The first four are $000011, 000101, 000110, 001001$. A prefix block can be counted by choosing remaining one positions.

2. Among strings beginning 00, those beginning 000 contribute $\binom{3}{2} = 3$, and the next prefix 001 contributes $\binom{3}{1} = 3$.

3. Rank five is the second member of the latter block: $001010_2 = 10$.

例题解析

1. 前四个为 $000011, 000101, 000110, 001001$；每个前缀块的大小可由剩余一的位置选择数求出。

2. 在以 00 开头的串中，前缀 000 贡献 $\binom{3}{2} = 3$，随后前缀 001 贡献 $\binom{3}{1} = 3$。

3. 第五名是后块第二个：$001010_2 = 10$。

Numerical order equals lexicographic order only after the length is fixed.

只有固定长度后，数值顺序才等于字典序。

Practice / 配套练习: AIME-2003-I-Q13 (WB p. 591), AIME-2012-II-Q07 (WB p. 587), AIME-2017-II-Q04 (WB p. 573), AIME-2004-II-Q10 (WB p. 736)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 6 / 方法 6 [NT06-M06]

Optimize a numeral by digit mass and lexicographic priority

按数位质量与字典序优先级优化整数

Procedure

1. Compare lengths first: every valid longer positive numeral exceeds every shorter one.

2. For a minimum with a prescribed digit sum, use the fewest positions and place as much mass as possible in low positions.

3. For a maximum with suffix constraints, maximize the earliest digit whose choice still permits a valid completion.

4. Use divisibility or geometric-progression constraints to certify that no rejected earlier prefix can work.

通用步骤

1. 先比较长度：每个有效的更长正整数都大于较短者。

2. 给定数位和求最小时，使用最少位数，并尽量把数位质量放在低位。

3. 带后缀限制求最大时，优先最大化仍允许合法补全的最早一位。

4. 用整除或等比限制证明被舍弃的更优前缀均不可能。

> Example. Find the smallest base-five numeral with digit sum 11.

> 例题：求数位和为 11 的最小五进制正整数。

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 6 / 方法 6 [NT06-M06] (continued / 续)

Worked solution

1. Two positions hold digit sum at most 8, so at least three positions are needed.

2. To minimize the leading digit, the two lower digits should total as much as possible, namely eight. The leading digit is therefore three.

3. The unique minimum is $344_5 = 99$; any larger length or earlier larger digit increases the value.

例题解析

1. 两位的数位和至多为 8，故至少需要三位。

2. 为最小化首位，低两位应合计最大值八，所以首位为三。

3. 唯一最小值为 $344_5 = 99$；更长位数或更大高位均会增大数值。

A greedy prefix choice needs a feasible completion, not just an attractive leading digit.

贪心选择前缀须有合法补全，不能只看首位大小。

Practice / 配套练习: AIME-2003-II-Q02 (WB p. 562), AIME-2009-I-Q01 (WB p. 566), AIME-2010-II-Q01 (WB p. 567), AIME-2015-II-Q03 (WB p. 571), AIME-2020-II-Q05 (WB p. 578)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 7 / 方法 7 [NT06-M07]

Model local digit rules by a directed transition graph

用有向转移图建模局部数位规则

Procedure

1. Use digits or the necessary recent-digit memory as states, and draw every allowed next-digit transition.

2. For a largest unrestricted-length numeral, check whether a reachable directed cycle permits arbitrarily long strings.

3. For a fixed length, count paths by a recurrence or by a uniform predecessor count; enforce the initial nonzero rule.

通用步骤

1. 以数字或所需最近数位记忆为状态，画出每个允许后继转移。

2. 位数不受限求最大数时，先检查可达有向循环是否允许任意长串。

3. 固定位数时，以递推或统一前驱数统计路径，并施加首位非零。

> Example. Count four-digit binary strings with no adjacent ones, allowing initial zeros.

> 例题：统计无相邻一的四位二进制串，允许前补零。

Worked solution

1. Use the last digit as the state. A zero permits either next digit; a one permits only zero.

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 7 / 方法 7 [NT06-M07] (continued / 续)

2. If a_n is the total count, final zero contributes a_{n-1} and final 01 contributes a_{n-2}. Thus $a_n = a_{n-1} + a_{n-2}$.

3. With $a_0 = 1, a_1 = 2$, the counts are $a_2 = 3, a_3 = 5, a_4 = 8$.

例题解析

1. 以末位为状态；零允许后接零或一，一只允许后接零。

2. 设总数为 a_n，以零结尾贡献 a_{n-1}，以 01 结尾贡献 a_{n-2}，故 $a_n = a_{n-1} + a_{n-2}$。

3. 初值 $a_0 = 1, a_1 = 2$，得 $a_2 = 3, a_3 = 5, a_4 = 8$。

Counting an acyclic longest path differs from counting all paths of a fixed length.

无环图的最长路径与固定长度全部路径是不同计数问题。

Practice / 配套练习: AIME-2001-II-Q01 (WB p. 559), AIME-2007-II-Q06 (WB p. 671)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 8 / 方法 8 [NT06-M08]

Prove uniqueness and carries in nonstandard positional systems

证明非标准位值系统的唯一性与进位

Procedure

1. Identify the legal digit range and the carry identity for the system.

2. Reduce modulo the smallest radix to determine the final digit, subtract it and divide to prove uniqueness inductively.

3. Use the carry rule to normalize signed digits or encode a combinatorial state, checking that supports do not overlap unexpectedly.

通用步骤

1. 确定系统的合法数位范围与进位恒等式。

2. 模最低进位基数确定末位，减去后相除，以归纳证明唯一性。

3. 用进位规则规范带符号数位或编码组合状态，并检查数位支撑不会意外重叠。

> Example. Write 8 in balanced ternary, with digits $-1, 0, 1$, and explain why the last digit is forced.

> 例题：把 8 写成数位为 $-1, 0, 1$ 的平衡三进制，并说明末位为何被迫确定。

Worked solution

1. Modulo three, $8 \equiv -1$, so its final balanced digit is -1.

2. Subtracting and dividing gives $(8+1)/3 = 3$. Its last digit is zero, followed by leading one.

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 8 / 方法 8 [NT06-M08] (continued / 续)

3. Thus $8 = 3^2 - 1$, with balanced digits $(1, 0, -1)$. Repeating the forced-residue argument proves uniqueness.

例题解析

1. 模三有 $8 \equiv -1$，故末位必为 -1。

2. 减去末位再相除得 $(8+1)/3 = 3$；它的末位为零，再往上为一。

3. 所以 $8 = 3^2 - 1$，平衡数位为 $(1, 0, -1)$；反复使用被迫剩余的论证可证明唯一性。

Uniqueness comes from legal digit ranges; an arbitrary signed expansion need not be unique.

唯一性来自合法数位范围，任意带符号展开未必唯一。

Factorial digits satisfy $0 \le f_i \le i$ because $(i+1)! = (i+1)i!$. The telescoping identity $a! - b! = \sum_{i=b}^{a-1} i\, i!$ produces a legal block of maximal digits; verify disjoint blocks before adding such expansions.

A set-count or prime-index state may encode each allowed element by a binary bit. Prove the update changes trailing one-bits to zero and the next zero to one; the claimed index then follows by binary uniqueness, not by guessing successive terms.

阶乘数位满足 $0 \le f_i \le i$，因为 $(i+1)! = (i+1)i!$。望远镜恒等式 $a! - b! = \sum_{i=b}^{a-1} i\, i!$ 产生一块合法最大数位；叠加展开之前须验证各块不相交。

集合计数或质数索引状态可用二进制位编码每个允许元素。须证明更新把末尾连续一变零、随后一个零变一；所求下标来自二进制唯一性，而非猜测逐项数值。

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 8 / 方法 8 [NT06-M08] (continued / 续)

Practice / 配套练习: AIME-2000-II-Q14 (WB p. 590), AIME-2008-II-Q04 (WB p. 565), AIME-2014-II-Q15 (WB p. 592), AIME-2024-II-Q06 (WB p. 642)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 9 / 方法 9 [NT06-M09]

Count digit patterns with leading-zero and multiplicity corrections

按首位零与重复数位修正数位模式计数

Procedure

1. Choose a pattern invariant: digit set, common digit sum, alternating order or palindrome half.

2. Count each pattern before substituting actual digit values, separating whether zero is present.

3. Apply bounded compositions or combinations only after checking that no digit exceeds its base limit; subtract repeated descriptions exactly once.

通用步骤

1. 选择模式不变量：数字集合、共同数位和、交替大小或回文半串。

2. 代入实际数字值前先数模式，并区分是否含零。

3. 使用有界拆分或组合前，先核对数位不超过底数限制；重复描述只消除一次。

> Example. How many two-digit decimal integers have digit sum 9 and distinct digits?

> 例题：有多少个两位十进制整数的数位和为 9 且两位不同?

Worked solution

1. Write $10a+b$ with $1 \le a \le 9$ and $b = 9 - a$.

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 9 / 方法 9 [NT06-M09] (continued / 续)

2. Every such a gives a valid digit b in $0, \ldots, 8$. Equality $a = b$ would require $2a = 9$, impossible.

3. There are nine values. Zero is permitted in the units place, so 90 is included.

例题解析

1. 写为 $10a + b$，其中 $1 \le a \le 9, b = 9 - a$。

2. 每个 a 都给出 $0, \ldots, 8$ 中有效数字 b；若 $a = b$ 则 $2a = 9$，不可能。

3. 共有九个值；个位允许零，故包括 90。

Leading-zero restrictions act on positions; repeated-digit restrictions act on values.

首位零限制针对位置，重复数位限制针对数值。

Practice / 配套练习: AIME-2003-I-Q09 (WB p. 649), AIME-2004-I-Q06 (WB p. 625), AIME-2004-II-Q04 (WB p. 614), AIME-2004-II-Q14 (WB p. 552), AIME-2013-I-Q02 (WB p. 570), AIME-2014-II-Q04 (WB p. 594), AIME-2024-II-Q03 (WB p. 581), AIME-2026-I-Q02 (WB p. 583)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 10 / 方法 10 [NT06-M10]

Count each digit uniformly by zero-padding a full block

补零成完整区块以均匀统计各数位

Procedure

1. Write all integers from 0 through $b^k - 1$ as strings of exactly k digits.

2. In any fixed position, each digit occurs exactly b^{k-1} times; sum a digit contribution position by position.

3. If leading zeroes have zero contribution they cause no change; otherwise correct them explicitly, and add or remove the interval endpoints.

通用步骤

1. 将 0 到 $b^k - 1$ 的整数全部写成恰有 k 位的数位串。

2. 每个固定位置上，各数字恰出现 b^{k-1} 次；逐位求数字贡献之和。

3. 若首补零贡献为零，则不改变结果；否则须明确修正，并增删区间端点。

> Example. Find the sum of the decimal digit sums of all integers from 0 through 999.

> 例题：求 0 到 999 各整数的十进制数位和之总和。

Worked solution

1. Pad every numeral to three digits; zero padding does not change its digit sum.

2. At each of three positions, every digit appears 100 times and the digits total 45.

3. The total is $3 \cdot 100 \cdot 45 = 13500$.

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 10 / 方法 10 [NT06-M10] (continued / 续)

例题解析

1. 每个数补为三位，补零不会改变数位和。

2. 三个位置的每一个位置上，每个数字出现 100 次，而所有数字之和为 45。

3. 所以总和为 $3 \cdot 100 \cdot 45 = 13500$。

For reciprocal-digit sums, define the contribution of zero separately rather than divide by zero.

倒数数位和须单独定义零的贡献，不能除以零。

Practice / 配套练习: AIME-2006-II-Q14 (WB p. 481)